

Political Statement Contradiction & Fact Checker

< Development Plan v2 — Contradiction + Fact-Check />

정치 발언 모순 탐지기 · 2-day hackathon · Journalism × AI track

Stack: Next.js + Claude API (Opus 4.8, web search) + Whisper STT + INSEE (FR) & KOSIS (KR) stats APIs

1. Overview & Vision

Across Korean, French, and other politics, cases where politicians **contradict their own past statements** or **make factually false claims** have become a major talking point — a leading driver of political distrust, especially among younger people. This project verifies political statements along **two independent axes**.

Two verification axes: ① **Self-consistency** — does it contradict the person's own past words (side-agnostic, fully neutral)? ② **Factuality (fact-check)** — is the claim actually true (cross-checked against official government statistics + web search)? The two axes are **kept separate with distinct labels** to preserve neutrality.

Vision vs. this hackathon's scope

- **Ultimate goal (roadmap):** a program that runs in **real time** during a live political broadcast, catching contradictions and falsehoods the moment a politician speaks.
- **This hackathon (48h) web demo:** after a broadcast, **upload the footage**, extract statements, then ① cross-check against past statements to detect contradictions and compute a **consistency %**, and ② **fact-check** the factual claims against official statistics + the web, showing true / false / unverifiable with sources.

Judging criteria mapping: ① real problem + numeric proof ② working core feature ③ AI-usage explanation. Fact-check directly addresses the panel journalists' key pain point — **verification in the AI era** →

aiming for Best Journalism.

2. Core Features (this demo)

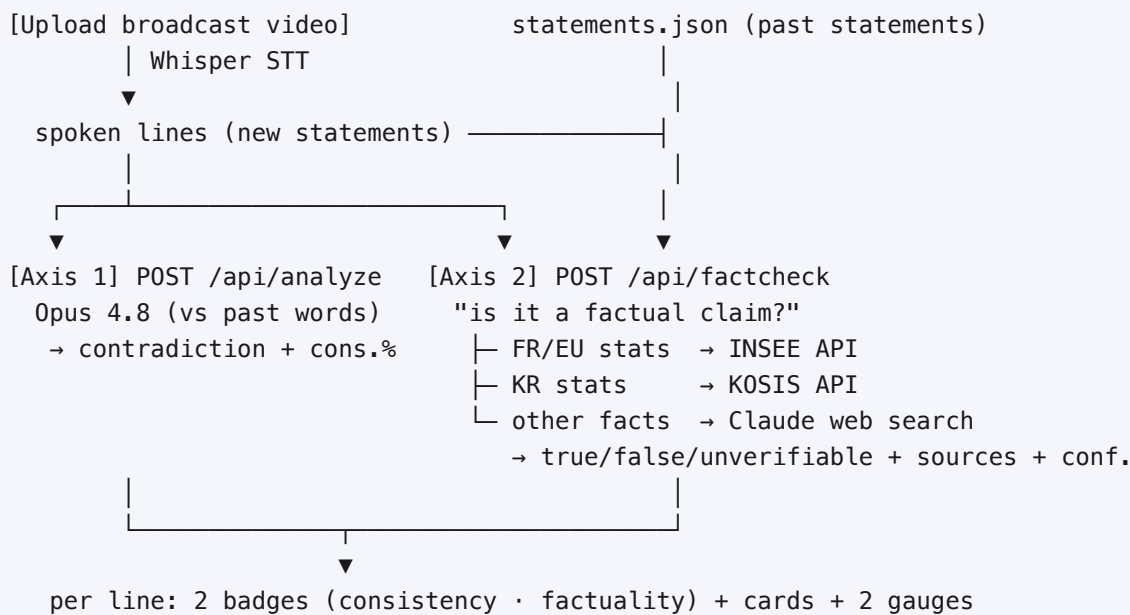
- 1. **Upload video** → extract statements (STT).
- 2. **[Axis 1] Contradiction detection** — cross-check each statement against the past-statements DB → contradiction verdict + reason, source, date.
- 3. **[Axis 2] Fact-check** — for checkable factual claims only, verify via **Claude web search + INSEE/KOSIS official statistics** → true / false / unverifiable + sources + confidence.
- 4. **Two scores:** consistency % = (non-contradicting / total)×100, accuracy % = (verified-true claims / checked claims)×100.
- 5. **Results dashboard:** two badges per statement (consistency & factuality) + cards + two score gauges.

3. Tech Stack

Layer	Choice	Notes
Frontend	Next.js (App Router) + TypeScript + Tailwind	Ships + deploys as one; polished demo
Backend	Next.js API Routes + <code>@anthropic-ai/sdk</code>	No separate server needed
STT (video→text)	OpenAI Whisper API	Korean supported. Claude has no STT. Fallback: upload transcript (.txt/.srt)
Contradiction + scoring	<code>claude-opus-4-8</code> · adaptive thinking · structured outputs	Fits reasoning-heavy judgment
Fact-check engine	<code>claude-opus-4-8</code> + web search server tool (<code>web_search_20260209</code>)	Live search + cited sources . General fact verification
Authoritative stats	INSEE (France) · KOSIS (Korea) OpenAPI	Verify statistical claims against official gov data. Free · API key
Data	<code>statements.json</code> (20-40 curated past statements)	No embeddings / vector DB — inject all into the 1M context
Deploy	Vercel (+ Ngrok for local demo)	Demo from one URL on stage

Extension roadmap (for the pitch): real-time streaming STT + live judgment, Supabase + Voyage AI embeddings, automated news / assembly API collection, more national stats sources.

4. Architecture & Data Flow



STT-risk fallback: feed the transcript straight into the APIs so the demo never breaks.

5. Data Model (TypeScript)

```

interface Statement {           // past statement (statements.json)
  id: string; politician: string; text: string;
  date: string; sourceUrl: string; topic: string;
}

interface SpokenLine { text: string; timestamp: number; }

interface ConsistencyVerdict { // Axis 1: contradiction
  line: string; isContradiction: boolean;
  pastStatement?: string; reason: string; confidence: number;
}

interface FactVerdict {        // Axis 2: factuality
  line: string;
  isFactualClaim: boolean;      // checkable factual claim?
  verdict: "TRUE" | "FALSE" | "UNVERIFIABLE";
  reason: string;
  sources: { title: string; url: string }[];
  sourceType: "INSEE" | "KOSIS" | "WEB";
  confidence: number;
}

interface AnalysisResult {
  politician: string;
  consistency: ConsistencyVerdict[];
  facts: FactVerdict[];
  consistencyScore: number;      // 0-100
  accuracyScore: number;        // 0-100
}
  
```

6. API Design

POST /api/transcribe — video/audio → Whisper → `SpokenLine[]`

POST /api/analyze — compare vs past words (contradiction + consistency %) via Opus 4.8 structured output

POST /api/factcheck — verify factual claims. Routing: statistical claims query INSEE/KOSIS as evidence; otherwise web search.

```
// app/api/factcheck/route.ts (core) — uses the web search server tool
import Anthropic from "@anthropic-ai/sdk";
const client = new Anthropic();

const res = await client.messages.create({
  model: "claude-opus-4-8",
  max_tokens: 8000,
  thinking: { type: "adaptive" },
  tools: [{ type: "web_search_20260209", name: "web_search" }], // live search +
  citations
  output_config: {
    format: { type: "json_schema", schema: {
      type: "object",
      properties: {
        facts: { type: "array", items: { type: "object", properties: {
          line: { type: "string" },
          isFactualClaim: { type: "boolean" },
          verdict: { type: "string", enum: ["TRUE", "FALSE", "UNVERIFIABLE"] },
          reason: { type: "string" },
          confidence: { type: "number" }
        }, required: ["line", "isFactualClaim", "verdict", "reason", "confidence"],
        additionalProperties: false }},
        accuracyScore: { type: "number" }
      }, required: ["facts", "accuracyScore"], additionalProperties: false }
    },
  },
  messages: [{ role: "user", content: buildFactPrompt(lines, statsContext) } ]
});
```

Note: On Opus 4.8, sampling params like `temperature` return a **400 error** — do not send them. The backend fetches INSEE/KOSIS first and passes it as `statsContext` (or wire them as custom tools).

Auth: `ANTHROPIC_API_KEY` · `OPENAI_API_KEY` · `INSEE_KEY` · `KOSIS_KEY` .

7. Prompt Design

[Fact-check prompt]

You are a careful fact-checker. Rules:

1) Check **ONLY** verifiable factual claims (numbers, statistics, historical facts).

Opinions/predictions/values ("a tax hike is right") → isFactualClaim=false, skip.

- 2) For statistical claims, use the provided official stats (INSEE/KOSIS) as the primary evidence; otherwise use web_search to find trustworthy sources.
- 3) Verdict is one of TRUE / FALSE / UNVERIFIABLE. Do not force true/false.
- 4) Attach a reason and sources to every verdict, and a confidence (0-1).
- 5) accuracyScore = (# TRUE / # checked factual claims) × 100.

[Provided official stats] {{stats_context}}
 [Statements to verify] {{spoken_lines}}
 Answer ONLY in the given JSON schema.

The consistency prompt is unchanged from v1 (same person only, differing context ≠ contradiction, source + confidence).

8. UI / UX

- **Flow:** select politician → upload video/transcript → transcription & analysis progress → results dashboard → warning states.
- **Two badges per statement:**  consistency (consistent/contradiction) ·  factuality (true/false/unverifiable). Click a card to expand reason, sources, and the past statement.
- **Two score gauges up top:** consistency % · accuracy %.
- **Design direction:** clean, neutral, trust-oriented. Always show source links so it reads as **quotation + verification**, not an accusation.

9. Two-Day Timeline (mornings = Web / afternoons = AI)

When	Work
Day 1 AM (Web)	Scaffold Next.js, upload UI + dashboard (2 badges, 2 gauges) skeleton, curate <code>statements.json</code>
Day 1 PM (AI)	Axis 1 contradiction <code>/api/analyze</code> + consistency % structured output; prompt engineering
Day 2 AM	Axis 2 fact-check <code>/api/factcheck</code> (web search tool) + INSEE/KOSIS integration, Whisper STT (+ transcript fallback), integration + UI polish
Day 2 PM	Deploy to Vercel, demo rehearsal, 5-min pitch + poster + timeline visual

10. Proving the Problem (research items)

- **Domestic:** KIPA government/assembly trust, Statistics Korea social survey.
- **Global:** OECD Trust in Government, Edelman Trust Barometer, Pew Research.

⚠️ **Note:** Verify the exact latest figures from primary sources before the pitch.

11. AI & External-API Disclosure (required)

- **STT:** Whisper (video → text).
- **Contradiction + scoring:** Claude Opus 4.8.
- **Fact-check:** Claude Opus 4.8 + web search server tool.
- **Authoritative stats:** INSEE (France), KOSIS (Korea) official government OpenAPIs.
- **Embeddings:** none. **Final verification:** human → "AI does the first pass, humans verify" (essential for fact-check).

12. Risks & Mitigations

Risk	Mitigation
Perceived political bias	Self-comparison (neutral) + fact-check strictly source-based ; keep the two axes separate
Fact-check error / defamation	Checkable claims only, TRUE/FALSE/ UNVERIFIABLE , sources always, human final check
STT accuracy	Transcript fallback + human correction
False positives	Context-aware prompt + confidence + human check
Stats-API limits	KOSIS dev account 1,000/day (enough for demo); web-search fallback when no stat exists
API cost / latency	Cap dataset size, streaming, caching

13. Demo & Pitch Deliverables

- Working demo: upload video → contradiction + fact-check + two scores.
- 5-min elevator pitch script / project visual-poster / catchy timeline.
- **Secure in advance:** 1-2 famous "flip-flop" clips + 1-2 **cleanly-checkable statistical claims** (ones INSEE/KOSIS resolve decisively).

14. Environment Setup

```
npx create-next-app@latest contradiction-detector --typescript --tailwind --app
cd contradiction-detector
npm i @anthropic-ai/sdk openai
```

```
# .env.local
ANTHROPIC_API_KEY=sk-ant-...
OPENAI_API_KEY=sk-...      # Whisper STT
INSEE_KEY=...              # register at portail-api.insee.fr
KOSIS_KEY=...              # register at kosis.kr/openapi

npm run dev                # http://localhost:3000
```

15. Verification (End-to-End)

1. Run `npm run dev`, upload a "flip-flop" clip + a transcript containing a statistical claim.
2. Axis 1: contradiction cards appear with date & source; consistency % computed.
3. Axis 2: statistical claims are judged true/false with an INSEE/KOSIS source, ambiguous ones marked unverifiable.
4. Differing-context statements / opinions are NOT falsely flagged or over-fact-checked (unverifiable / N/A).
5. The deployed Vercel URL behaves the same; if it fails, fall back to Ngrok.